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# Semantic Warp: Provenance-Conditioned Completion in Collective Memory

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## Abstract

**Problem.** In collective memory for multi-agent navigation, an agent can commit to — and reach — a target it has never observed: all supporting evidence arrived from peers. Folk culture calls this navigating by someone else’s song; we call it a *semantic warp*. Success-only and even stage-level metrics cannot see it, because no existing evaluation conditions completion on the *provenance* of the evidence behind the commitment. **Approach.** We define the warp event  $W_\theta^* := M^* \wedge (\varphi \geq \theta)$ , where foreignness  $\varphi$  is the peer share of the evidence mass *in the same weights the merge used to select the candidate*, frozen at lock time. Warp is not a fifth Q/R/M/C stage but an annotation that splits  $P(C^*|M^*)$  into provenance strata. Because symbolic merges decompose by source, we obtain (i) a counterfactual attribution of warp value via deterministic replay with foreign evidence masked, (ii) a closed-form *warp distance law* from trust $\times$ staleness rules, and (iii) a decentralised *Warp Drive* protocol (optimistic lock + reservation + anti- $M^*$  rollback) for the warp-collision pathology. **Findings.** On a 2,400-episode scarcity sweep, raw warp locks almost never complete directly ( $P(C^*|W^*) = 0.004\text{--}0.020$  vs.  $0.22\text{--}0.41$  for own locks; CIs disjoint), yet foreign evidence still carries value as transport: up to 51% of own-lock completions are warp-assisted chains. The distance law predicts intention-formation horizons tick-exactly, including a 6/6-exact hold-out with predictions registered before execution. Warp Drive turns the trade-off around: success rate rises at every cadence (disjoint CIs), and fast sharing with the protocol beats the best fixed-cadence baseline ( $0.614\text{--}0.620$  vs.  $0.583$ ). A cross-session, cross-agent strict warp is replicated on an LLM substrate (10/10 layouts vs. 0/10 without foreign memory). **Thesis.** The value of collective memory lies less in the information itself than in the *coordination contract attached to the information*. **Scope.** Grid-world substrates and one small text-world LLM demo; the contribution is a measurement layer, a falsifiable law, and a minimal protocol, not a general theory.

## 1 Introduction

In songline cultures, a traveller who knows the song can cross terrain they have never seen: navigational competence transfers without physical visitation. Multi-agent systems with shared memory perform the same act constantly — an agent commits to a water source, a goal region, or a tool location known only from a peer’s broadcast — yet no standard evaluation distinguishes this act from ordinary memory-driven navigation. Success metrics aggregate over it; stage-level diagnostics such as Q/R/M/C count the commitment ( $M^*$ ) and the completion ( $C^*$ ) but not *whose evidence* backed them.

36 This paper makes that act a first-class measured event. A *semantic warp*  $W^*$  is a target materialisation  
 37 whose supporting evidence mass is (almost) entirely foreign, and *warp completion*  $P(C^*|W^*)$  asks  
 38 whether someone else’s song actually gets you there. Three properties of symbolic collective memory  
 39 make the construction sharp, and none of them holds for opaque vector memories:

- 40 1. **Merges decompose by source.** Foreignness  $\varphi \in [0, 1]$  is computed from the same per-source  
 41 weights (trust, staleness, support) the merge itself used to rank the candidate — it is a reading of  
 42 the decision, not a new heuristic.
- 43 2. **Replays are deterministic.** Re-running an episode with foreign contributions masked at merge  
 44 yields a *counterfactual* warp gain per episode, not a correlation.
- 45 3. **Sharing rules have closed forms.** Trust and staleness decay imply a *warp distance law*: an  
 46 explicit horizon beyond which foreign evidence can no longer form intention — falsifiable, and  
 47 falsifiably different across architectures.

48 Our measurements then force a reframing. Raw warp is nearly worthless at the point of completion:  
 49 under scarcity, warp locks complete an order of magnitude less often than own-evidence locks  
 50 (§4), because concurrent recipients of the same broadcast race for the same target and most must  
 51 lose. Yet the same foreign evidence is demonstrably valuable as *transport*: up to half of all own-  
 52 evidence completions are chains that began with a failed warp lock which carried the agent into  
 53 self-confirmation range (§4.1). And a lightweight decentralised contract — a reservation piggybacked  
 54 on the existing broadcast plus an anti- $M^*$  rollback in the sense of Time Warp simulation Jefferson  
 55 (1985) — recovers what raw warp loses: with the protocol, *fast* sharing becomes the best-performing  
 56 regime on unconditional success rate (§6). Hence the thesis in one sentence: **the value of collective**  
 57 **memory is not the information; it is the coordination contract attached to the information.**

## 58 Contributions.

- 59 1. The first provenance-conditioned completion metric for collective memory:  $W_\theta^*$ ,  $\varphi$   
 60 frozen at lock time, the decomposition  $P(C^*|M^*) = P(C^*|M^*, \varphi < \theta) P(\varphi < \theta|M^*) +$   
 61  $P(C^*|W^*) P(W^*|M^*)$ , and the counterfactual warp gain via source-masked deterministic  
 62 replay (§3–§4).
- 63 2. The *warp distance law*  $\text{age}_{\max}(\tau_i) = \alpha^{-1} \ln(\tau_i c / \tau)$  derived from the memory’s own  
 64 trust×staleness rules, validated tick-exactly including a registered-prediction hold-out (6/6  
 65 exact, one cell where a pruning horizon must undercut the exponential gate — and does) and an  
 66 architecture-level shape difference: fixed-cadence peer memory has *no* gate (§5).
- 67 3. The *Warp Drive* protocol: optimistic warp lock, reservations carried on the existing broadcast  
 68 cadence (no coordinator), and anti- $M^*$  rollback with exponential backoff. It resolves the  
 69 known misled-agent pathology preventively, bounds rollback latency by  $\approx K$  ticks, and lifts  
 70 unconditional success rate above the best fixed-cadence baseline (§6).
- 71 4. A cross-session, cross-agent strict warp on an LLM substrate: an agent that has never observed  
 72 a room solves a task there purely on another agent’s previous-session memory, replicated 10/10  
 73 layouts against a 0/10 control (§7).

74 **Relation to the companion papers.** The Q/R/M/C framework paper treats stage rates as the  
 75 measurement instrument; the collective-memory paper establishes the architectures and the cadence  
 76 trade-off. This paper adds the provenance axis on top of both: the same logs, one new annotation,  
 77 and the mechanisms the annotation exposes.

## 78 2 Setting

79 **Substrate and task.**  $N$  agents explore a  $12 \times 10$  grid with  $M$  semantic water targets ( $N > M$ :  
 80 scarcity) and hazards. Each agent maintains symbolic place memory (tags, confidences, support,  
 81 freshness). Four sharing architectures over the identical substrate and planner: independent (no  
 82 communication), shared (single event bus), centralized (consensus aggregator), peer (broadcast  
 83 every  $K$  ticks, asymmetric trust), plus CSM (peer + explicit merge/trust/staleness rules with decay  
 84 rate  $\alpha=0.05$  and inclusion threshold  $\tau=0.30$ ).

85 **Q/R/M/C instrumentation.** Per tick and per agent we log query formation (Q), retrieval satisfaction  
 86 (R), target materialisation as the planner’s lock ( $M^*$ ), and completion ( $C^*$ ). Episode-level conditional  
 87 rates follow the companion definition. All warp measurements below are annotations of these  $M^*$   
 88 events; no planner or memory invariant is modified.

### 89 3 The warp event

90 **Provenance and foreignness.** Every evidence unit entering a merge carries its source. For the  
 91 candidate  $m$  the planner locks,

$$\varphi(m) = \frac{\sum_{e \in \text{supp}(m), \pi(e) \neq \text{self}} w(e)}{\sum_{e \in \text{supp}(m)} w(e)},$$

92 where  $w(e)$  are *the merge’s own weights*:  $\text{trust} \cdot \log(1 + \text{support})$  for peer/centralized consensus,  
 93  $\text{trust} \cdot e^{-\alpha \text{age}} \cdot \text{conf}$  for CSM, observation masses for the shared bus. For peer and centralized  
 94 architectures  $\varphi$  is reconstructed exactly from the per-contribution fields the merge already exports;  
 95 the measurement layer reads state and never writes it.

96 **Freezing rule.**  $\varphi$  is fixed at the moment of the lock. If the agent later confirms the place with its  
 97 own eyes,  $\varphi$  is *not* recomputed — otherwise every successful warp would dilute itself on approach  
 98 and  $P(C^*|W^*)$  would be unmeasurable by construction.

99 **Warp events and strata.**  $W_\theta^* := M^* \wedge (\varphi \geq \theta)$  with  $\theta=1$  (strict: never self-observed, cross-  
 100 checked against an observation ledger) and  $\theta=0.8$  (soft). Warp share  $P(W^*|M^*)$  is a structural  
 101 descriptor (identically 0 for independent); warp completion  $P(C^*|W^*)$  is the new quantity. The  
 102 chain-rule decomposition in §1 makes warp an exact refinement of  $P(C^*|M^*)$ , not a new stage:  
 103 identifiability of Q/R/M/C is untouched.

104 **Regression anchor.** The known misled-agent episode of the architecture comparison (agent A locks  
 105 (3, 8) known only from C; C claims it first; A strands) is annotated by the instrumentation as  $\varphi = 1.0$   
 106 strict  $W^*$  with collision — the pathology the framework must see, and the protocol of §6 must fix.

### 107 4 Raw warp rarely completes — but transports

108 **Design.** 2,400 episodes:  $(N, M) \in \{(3, 2), (5, 3), (8, 5), (4, 4), (6, 2)\} \times \text{three layouts} \times$   
 109  $\{\text{peer-}K2, \text{peer-}K8, \text{CSM-}K8, \text{shared}\} \times 20 \text{ seeds} \times \{\text{full, foreign-masked}\}$  with identical seeds.  
 110 Unit of analysis: lock event; CIs by episode-cluster bootstrap.

111 Figure 1 shows the cadence view on a dedicated peer sweep ( $K \in \{1, 2, 4, 8, 16\}$ ): warp share is  
 112 what fast sharing buys, and it is precisely the stratum that does not complete. Table 1 gives the  
 113 architecture view.

Table 1: Provenance strata of completion under scarcity ( $N > M$ ), full runs. Warp locks complete an order of magnitude less often than own-evidence locks; all CI pairs are disjoint. (H-W1 confirmed.)

| Architecture | $P(C^* W_{0.8}^*)$   | $n_{W^*}$ | $P(C^* M^*, \text{own})$ | $n_{\text{own}}$ |
|--------------|----------------------|-----------|--------------------------|------------------|
| peer- $K=2$  | 0.004 [0.001, 0.008] | 1,437     | 0.223 [0.186, 0.275]     | 2,738            |
| peer- $K=8$  | 0.020 [0.000, 0.038] | 149       | 0.389 [0.329, 0.446]     | 1,734            |
| CSM- $K=8$   | 0.017 [0.006, 0.044] | 637       | 0.414 [0.368, 0.453]     | 1,684            |
| shared       | 0.004 [0.002, 0.008] | 2,099     | 0.279 [0.237, 0.332]     | 2,250            |

#### 114 4.1 Warp as transport: lock chains

115 Direct completion understates the value of foreign evidence. A *warp-assisted own completion* is  
 116 a completed own-evidence lock preceded (same agent, same episode) by a soft- $W^*$  lock that was  
 117 dropped: the warp failed as a destination but succeeded as a ride.

118 This resolves an apparent paradox in the counterfactual data: the shared bus at moderate scarcity  
 119 gains +10.8 ticks [+2.2, +20.9] from foreign evidence although its direct  $P(C^*|W^*)$  is 0.004. The

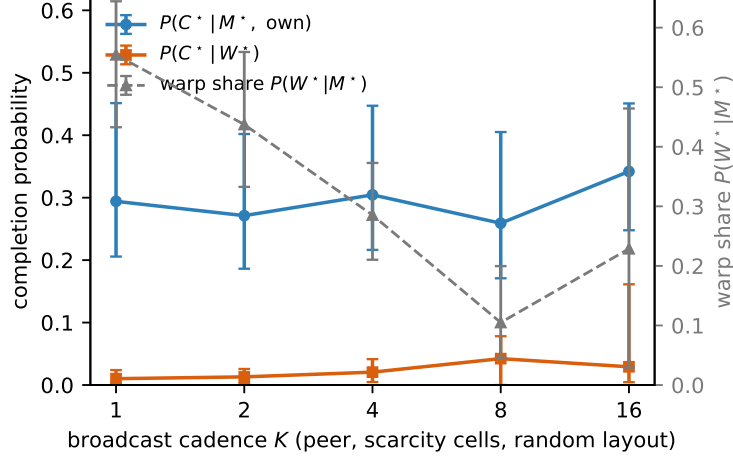


Figure 1:  $\varphi$ -stratified completion vs. broadcast cadence  $K$  (dedicated peer sweep: scarcity cells, random layouts, 20 seeds, episode-cluster bootstrap CIs). As  $K$  shrinks, the warp share  $P(W^*|M^*)$  grows from 0.11 to 0.55 while  $P(C^*|W^*)$  stays below 0.05 and the own-stratum rate stays flat: the completion collapse of fast sharing is concentrated in the  $W^*$  stratum, giving the companion paper’s bottleneck shift its mechanistic reading. The share uptick at  $K=16$  (broadcasts are rare, but agents also discover little on their own) is reported as observed.

Table 2: Lock-chain analysis (random layouts, full runs). The first  $W^*$  lock in a chain is made from  $\approx 8$ –10 cells away; the completing own lock from 2.0 cells — exactly the observation radius, i.e. the self-confirmation point.

| Arm         | own $C^*$ | warp-assisted (share) | $r(\text{first } W^*) \rightarrow r(\text{own})$ |
|-------------|-----------|-----------------------|--------------------------------------------------|
| shared      | 248       | 127 ( <b>51.2%</b> )  | 9.5 $\rightarrow$ 2.0                            |
| peer- $K=2$ | 223       | 99 ( <b>44.4%</b> )   | 9.7 $\rightarrow$ 2.0                            |
| CSM- $K=8$  | 279       | 13 (4.7%)             | 5.8 $\rightarrow$ 2.0                            |
| peer- $K=8$ | 246       | 7 (2.8%)              | 8.4 $\rightarrow$ 2.0                            |

gain flows through chains; the freezing rule books the completion honestly in the own stratum, and the chain analysis recovers the warp’s contribution as a separate measured quantity.

## 4.2 Counterfactual warp gain

Masking foreign evidence at merge under the same seed gives  $WG = t_{\text{cens}}(\text{masked}) - t_{\text{cens}}(\text{full})$  per episode. The sign map over (layout, scarcity  $\rho=N/M$ ) is regime-dependent: positive for the shared bus at unpredictable layouts and moderate scarcity (+10.8 at  $\rho=1.5$ ), negative for fast peer sharing (peer- $K=2$  random:  $-7.1 [-13.5, -1.3]$ ),  $\approx 0$  at hard scarcity. A per-agent mask (one agent masked, others share normally) confirms the sign at the individual level:  $WG_i = -10.25 [-20.91, -0.16]$  ticks on the fast-share scarcity cell, with spillover on unmasked agents  $\approx 0$  ( $-0.62 [-3.06, +1.91]$ ). Raw warp *hurts* its recipient precisely where sharing is fastest — the collision cost exceeds the information value.

**Exploratory: collision pressure.** Co-lock pressure (other agents locked on the same target) correlates with warp failure with the predicted sign but negligible magnitude (Spearman +0.033,  $p=0.018$ ,  $n=5,117$ ); we report it as exploratory rather than as a confirmed predictor.

## 5 The warp distance law

CSM includes foreign evidence with weight  $\tau_i e^{-\alpha \text{age}} c$  (trust  $\tau_i$ , confidence  $c=0.95$ ) and inclusion threshold  $\tau=0.30$ . Foreign evidence therefore stops forming intention at

$$\text{age}_{\max}(\tau_i) = \frac{1}{\alpha} \ln(\tau_i c / \tau), \quad \text{defined for } \tau_i c > \tau.$$

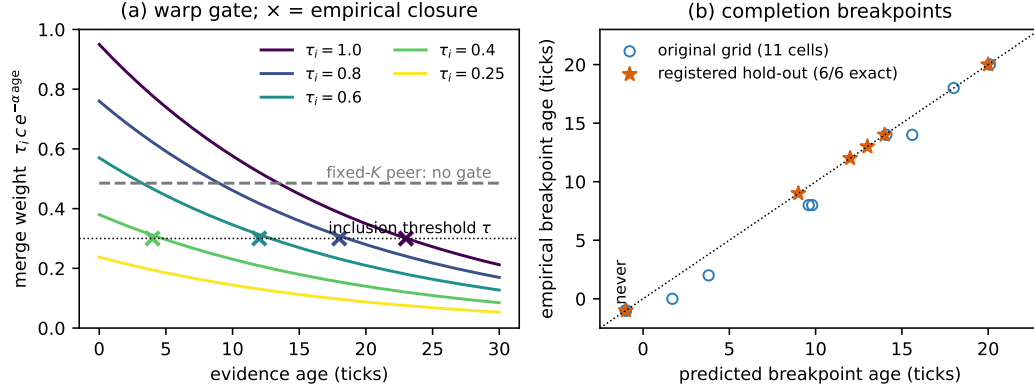


Figure 2: The warp distance law. **(a)** Foreign-evidence merge weight under trust  $\times$  staleness decay;  $\times$  marks the empirically measured last age at which the traveller’s query still exposes the target — each sits on its curve’s crossing of the inclusion threshold  $\tau$ . Fixed-cadence peer memory has no staleness term (dashed): its gate never closes; the architectures differ in the *shape* of the curve. **(b)** Predicted vs. empirical completion breakpoints in full navigation episodes: original grid (circles; age sampled in steps of 2, hence points at most one grid step below the diagonal) and the registered hold-out (stars; age step 1, predictions written to disk before execution, 6/6 exact, including the never-cell at  $(-1, -1)$  and the cell where the pruning horizon undercuts the exponential gate).

137 **Witness–traveller protocol.** A witness observes water once and goes silent; a traveller who has  
 138 never seen it receives the snapshot at a parameterised age. Every lock the traveller makes is a strict  
 139  $W^*$  by construction. Figure 2 summarises both experiments.

Table 3: Gate curve: largest evidence age at which the traveller’s query still exposes the target. Fixed-cadence peer memory has no staleness term: the gate never closes (checked to age 120 at all trusts) — the architectures differ in the *shape* of the curve, not just its level.

| trust $\tau_i$ | empirical age <sub>max</sub> | predicted      |
|----------------|------------------------------|----------------|
| 1.00           | 23                           | 23.05          |
| 0.80           | 18                           | 18.59          |
| 0.60           | 12                           | 12.84          |
| 0.40           | 4                            | 4.73           |
| 0.25           | gate closed                  | 0 (trust-flip) |

140 **Completion breakpoint.** In full navigation episodes the feasibility inequality acquires one discrete  
 141 refinement: the traveller self-confirms the target once it is within the observation radius, so the foreign  
 142 gate only has to survive until tick  $t_{\text{gate}} = d - r_{\text{obs}} - 1$ , giving breakpoint  $\text{bp} = \text{age}_{\text{max}} - t_{\text{gate}}$ ;  
 143 snapshot pruning executes on broadcast ticks and adds the bound  $10K - t_{\text{bcst}}$ . All constants of  
 144 the law are unchanged. All 11 CSM cells of the original grid match within sampling resolution;  
 145 completion curves are monotone steps; the trust-flip zeroes  $W^*$  events discretely.

Table 4: Registered-prediction hold-out: six cells absent from the original grid, predictions written to disk *before* any episode, age grid step 1. All six are exact integer matches, including the never-cell and the cell where the pruning horizon must undercut the exponential gate.

| Cell                    | binding bound  | predicted | empirical    |
|-------------------------|----------------|-----------|--------------|
| $K=8, \tau_i=0.9, d=9$  | gate           | 14        | <b>14</b>    |
| $K=8, \tau_i=0.7, d=9$  | gate           | 9         | <b>9</b>     |
| $K=8, \tau_i=0.5, d=15$ | gate           | never     | <b>never</b> |
| $K=4, \tau_i=1.0, d=6$  | gate           | 20        | <b>20</b>    |
| $K=2, \tau_i=1.0, d=12$ | <i>pruning</i> | 12        | <b>12</b>    |
| $K=2, \tau_i=0.8, d=8$  | gate           | 13        | <b>13</b>    |

The law is thus predictive out of sample, tick-exactly: the CSM rules really are the mechanism that forms (and un-forms) intention. It also reinterprets the companion result that CSM dominates fixed-cadence peers: trust $\times$ staleness is a *warp gate* — CSM does not share better, it *refuses to warp on a stale or untrusted song*.

**Portability to continuous dynamics.** All three registered predictions survive on a continuous VMAS substrate through the same continuous $\rightarrow$ grid bridge used by the companion’s portability test: (i) the provenance strata stay an order of magnitude apart ( $P(C^*|W^*) = 0.012 [0.000, 0.042]$  vs. 0.562 [0.457, 0.709] at  $K=1$ ); (ii) warp share falls monotonically with cadence ( $0.723 \rightarrow 0.366 \rightarrow 0.000$  over  $K=1/4/16$ ); (iii) the law becomes *metric*: with the travel-to-sight time calibrated once on a fresh-evidence episode, the predicted breakpoints match within the sampling grid, and all cells beyond the metric warp radius ( $v \cdot \text{age}_{\max}$  shorter than the distance) are predicted *never* and observed *never*. One genuinely continuous finding: inertia is a physical warp-leak channel — an agent can coast into sighting range after the gate closes — removed by a no-lock-then-brake controller that restores the grid semantics.

## 6 The Warp Drive protocol

Warp collision — locking on foreign evidence for a target someone else will claim — is the dominant failure of fast sharing. Warp Drive adds three primitives, all decentralised:

1. **Optimistic warp lock** — locks on foreign evidence remain allowed immediately (planner unchanged).
2. **Warp reservation** — at the owner’s next broadcast tick the reservation rides on the ordinary snapshot (same cadence  $K$ , no coordinator, no extra channel); receivers suppress reserved targets in their queries. Priority: (lock tick, distance at lock, agent id).
3. **Anti- $M^*$  rollback** (Time Warp semantics Jefferson (1985)) — an agent holding a lock dominated by a foreign reservation retracts it: the planner re-queries, the target enters exponential backoff ( $4 \rightarrow 8 \rightarrow \dots \rightarrow 32$  ticks) with an agent-id tie-break against oscillation. The event log stays append-only.

**Decentralisation check.** Driving the protocol by direct per-agent calls — no runtime, no environment — produces the same retraction decisions from each agent’s own inbox alone.

**Misled-A anchor.** With Warp Drive the pathological lock is resolved *preventively*: C’s reservation arrives before A ever locks (3, 8); A completes on another target. In the sweep, rollback latency of dominated locks is finite and  $\approx K$ : 1.0/1.5/2.7 ticks at  $K=1/2/4$  (baseline: until end of episode).

Table 5: Unconditional results, 840 episodes (scarcity cells  $\times$  random/asymmetric layouts  $\times$  20 seeds). Success rate rises under Warp Drive at every cadence with disjoint CIs, censored time falls, and fast sharing with the protocol beats the  $K=8$  sweet-spot baseline (Figure 3). Lock census shows the conditional-metric denominator shrinking — which is why the claim is stated on unconditional quantities.

| Arm        | success rate                | $t_{\text{cens}}$ (fail = 120) | lock census |
|------------|-----------------------------|--------------------------------|-------------|
| $K=1$ base | 0.592 [0.574, 0.608]        | 55.3 [53.7, 57.2]              | 0.986       |
| $K=1$ +WD  | <b>0.620</b> [0.609, 0.629] | 53.1 [52.0, 54.4]              | 0.985       |
| $K=2$ base | 0.572 [0.549, 0.592]        | 57.2 [55.1, 59.7]              | 0.973       |
| $K=2$ +WD  | <b>0.614</b> [0.601, 0.624] | <b>53.2</b> [52.0, 54.6]       | 0.926       |
| $K=4$ base | 0.577 [0.554, 0.598]        | 57.3 [55.1, 59.8]              | 0.944       |
| $K=4$ +WD  | <b>0.620</b> [0.609, 0.630] | <b>52.2</b> [50.9, 53.7]       | 0.739       |
| $K=8$ base | 0.583 [0.558, 0.604]        | 55.2 [52.9, 57.9]              | 0.682       |

On the conditional axis, recovery is measured against the scarcity ceiling  $\min(m, M)/m$  (no protocol conjures extra water): 0.90/2.15/6.6 of the recoverable collapse at  $K=1/2/4$ ; values above 1 arise because reservation suppression removes doomed locks from the  $M^*$  population. Together with Table 2 this closes the arc: raw warp fails at completion (§4); with a coordination contract attached, warp at fast cadence becomes the best regime. The cadence sweet spot of the companion paper is

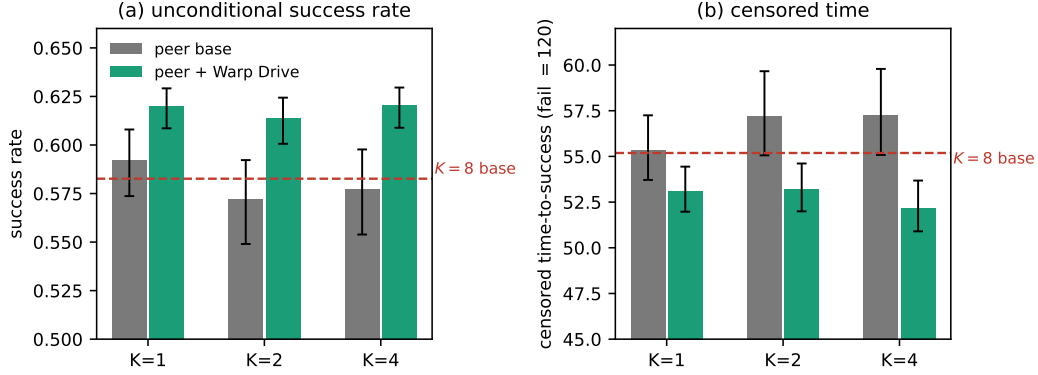


Figure 3: Warp Drive on unconditional metrics (840 episodes; bootstrap CIs). **(a)** Success rate rises at every cadence with disjoint CIs, and every fast- $K$  protocol arm exceeds the  $K=8$  sweet-spot baseline (dashed). **(b)** Censored time-to-success falls correspondingly. The claim that the protocol removes the main argument against fast sharing rests on these unconditional quantities, not on the (denominator-sensitive) conditional rates.

reinterpreted: moderate  $K$  was never about information freshness — it was an implicit rate limiter on warp collisions, and an explicit contract does the job better.

## 7 Cross-session warp on an LLM substrate

Finally, the strict warp crosses both an agent boundary and a session boundary on a language substrate. In session 1, agent A (a local LLM tag extractor over natural-language observations; Ollama llama3.1, deterministic caching, zero API cost) sweeps a three-room text household and its symbolic memory is dumped to disk. In session 2, agent B — a fresh agent in a fresh environment instance, with the stock directional oracle removed so that long-range knowledge can only come from memory — loads A’s dump as foreign provenance and solves “bring the apple to the table.” Every lock B makes on the apple or the table is  $\varphi=1.0$  strict  $W^*$ ; execution is layered (the LLM makes the semantic decisions — which remembered place to pursue, when to take/put — while locomotion to the locked waypoint is a deterministic motor primitive, matching the substrate’s planner stack).

Table 6: Cross-session warp over 10 randomised layouts. Completion is attributed at the environment’s affordance radius (take/put act within Manhattan distance 1, and the witness tags anchors on within-reach cells).

|                        | success      | mean $t_{\text{succ}}$ | layouts with completed strict $W^*$ |
|------------------------|--------------|------------------------|-------------------------------------|
| warp (A’s memory)      | <b>10/10</b> | 8.3                    | <b>10/10</b>                        |
| control (empty memory) | 0/10         | —                      | —                                   |

To our knowledge this is the first documented cross-session, cross-agent strict semantic warp on an LLM substrate. The 10/10 vs. 0/10 contrast shows the warp is load-bearing, not decorative.

**Scaling to a live collective.** With  $N=3$  LLM agents sharing by peer broadcast under scarcity ( $M=2$  apples, four layouts,  $K \in \{2, 8\}$ ), the full warp phenomenology reproduces on the language substrate (all predictions registered before the runs): strict  $W^*$  locks occur; warp share falls with cadence (0.417 at  $K=2$  vs. 0.083 at  $K=8$  — the same shape as the grid  $0.554 \rightarrow 0.105$  and VMAS  $0.723 \rightarrow 0.000$ ); warp collisions appear in 7/8 episodes. A paired comparison against the no-communication baseline on identical layouts shows raw sharing *lowering* success from 2/3 to 1/3 per episode — the collision cost exceeds the information value on the LLM substrate exactly as in §4.2. Attaching the unchanged Warp Drive protocol to the LLM collective closes the arc on language too: pooled success rises from 0.333 to 0.417 at both cadences with rollback latencies again  $\approx K$  (1 tick at  $K=2$ , 7 at  $K=8$ ). The recovery is partial ( $\approx 25\%$  of the collapse, vs. 90–100% on the grid): on a language substrate the collision pathology is compounded by semantic-level losses (hallucinated anchor tags, noisy locks)

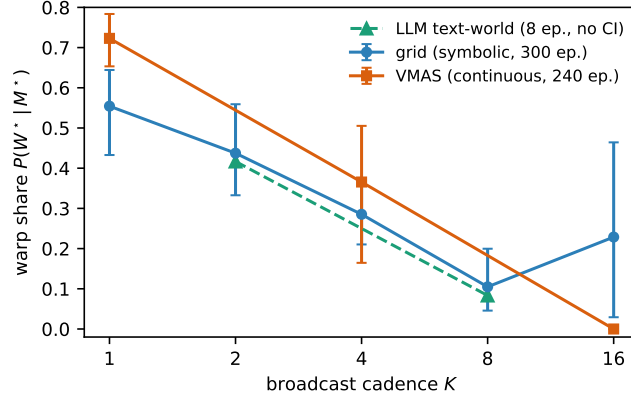


Figure 4: One quantity, three substrates: warp share  $P(W^*|M^*)$  vs. broadcast cadence  $K$  on the symbolic grid, the continuous VMAS scenario, and the LLM text-world. The curves nearly coincide — how much of a collective’s materialisation runs on foreign evidence is set by the sharing cadence, not by the substrate. (LLM arm: 8 episodes, point estimates only.)

207 that no reservation can cure — a measurable decomposition we leave to future work. The distance  
 208 law also survives LLM-produced evidence: with the gate constant  $c$  set to the confidence the extractor  
 209 actually emitted ( $c_{\text{meas}}=0.900$ ), all four registered breakpoints land within one tick of prediction.

210 Two interface findings transfer to any LLM-over-symbolic-memory design: the LLM query former  
 211 free-generates retrieval tags outside the memory’s vocabulary (requiring an any-of gate), and small  
 212 local models are reliable at semantic commitment but not at coordinate arithmetic or systematic  
 213 exploration (requiring the layered execution above and an explicit exploration tier — without one,  
 214 discovery never happens and the warp has nothing to feed on).

215 Figure 4 overlays the warp-share-cadence curve across all three substrates: the quantity behaves as a  
 216 substrate-free property of the sharing architecture.

## 217 8 Related work

218 **Shared memory in agent collectives.** Generative Agents Park et al. (2023) and multi-agent  
 219 LLM frameworks share memory streams or context but expose no lock interface and no per-source  
 220 decomposition of a merged belief, so provenance-conditioned completion is not measurable there.  
 221 Vector memories (e.g. MemGPT Packer et al. (2023)) merge by embedding arithmetic; the merge is  
 222 not source-decomposable, which blocks both  $\varphi$  and the masked-replay counterfactual.

223 **Emergent communication.** CommNet Sukhbaatar et al. (2016) and TarMAC Das et al. (2019)  
 224 learn continuous channels; the companion paper shows such channels collapse stage observability.  
 225 Warp requires an explicit materialisation event, which these architectures do not expose.

226 **Stigmergy.** Pheromone-style coordination is an *anonymous* warp: evidence transfers without  
 227 provenance. Our results quantify what anonymity costs (shared-bus warp share is maximal while its  
 228 warp completion is minimal) and what provenance buys (trust gates, reservations, rollback).

229 **Optimistic concurrency.** The rollback primitive is Time Warp Jefferson (1985) transplanted from  
 230 parallel simulation to memory-driven navigation: locks are optimistic computations, occupancy  
 231 evidence is a straggler message, anti- $M^*$  is the anti-message. To our knowledge this transplant is  
 232 new.

## 233 9 Limitations and honest negatives

234 (0) All experiments assume a SHARED coordinate frame: place identity is coordinate proximity with  
 235 semantics as a tie-breaker inside a spatial radius; the cross-agent place-correspondence problem is



236 bracketed, not solved. (A frame-free extension — fingerprint-based semantic identity with landmark-  
237 consensus frame recovery, failing closed under perceptual aliasing where the coordinate assumption  
238 fails open — is implemented and validated separately and is out of scope here.) (1) Grid worlds,  
239 a continuous VMAS scenario, and one small text world; no photorealistic substrate. (2) The  
240 co-lock-pressure predictor survived only as exploratory (§4.2). (3) The warp-gain sign map is  
241 regime-dependent rather than the clean interior structure the pre-registered hypothesis expected;  
242 deterministic symmetric/asymmetric layouts carry little seed variance, so the statistical weight rests  
243 on random layouts. (4) The distance-law refinement (self-confirmation horizon, pruning schedule)  
244 was formulated after seeing the original grid — mitigated, but not erased, by the registered hold-out.  
245 (5) The LLM demo uses one model family and a deliberately small environment; it is a demonstration  
246 of measurability, not a benchmark. (6) All deviations from the pre-registered design are listed in  
247 Appendix A.

## 248 10 Conclusion

249 Warp turns “agents share memory” from an architecture property into a measured, causally attributable  
250 event with a falsifiable distance law and a protocol that acts on it. The measurements dissolve the  
251 information-centric reading of collective memory: raw foreign evidence mostly fails its recipient at  
252 completion and pays off as transport, while a two-line coordination contract — reserve on lock, roll  
253 back on domination — makes the fastest sharing regime also the best one. Provenance, not volume,  
254 is the control surface that matters.

### 255 A Deviations from the pre-registered design

- 256 1. Reservation tie-break extended to (lock tick, *distance at lock*, agent id): a pure id tie-break hands  
257 simultaneous locks to a distant agent over the target’s near-discoverer.
- 258 2. Protocol recovery is measured against the scarcity ceiling  $\min(m, M)/m$  rather than the  $K=8$   
259 reference, separating structural from coordination collapse; unconditional metrics are reported  
260 alongside (Table 5).
- 261 3. Completion breakpoints use the discrete-time refinement of §5 (law constants unchanged;  
262 hold-out registered).
- 263 4. Collision pressure is operationalised as co-lock count rather than co-recipients (structurally  
264 constant  $N-2$  under broadcast).
- 265 5. LLM demo: layered execution (LLM semantics, deterministic locomotion); completion at-  
266 tributed at the affordance radius.

### 267 B Reproducibility

268 All experiments run on CPU in minutes-to-tens-of-minutes; the LLM demo uses a local Ollama  
269 model with deterministic caching.

```

270 PYTHONPATH=. python experiments/warp/exp_warp_anchor.py
271 PYTHONPATH=. python experiments/warp/exp_warp_gain.py --seeds 20
272 PYTHONPATH=. python experiments/warp/analyze_warp_chains.py
273 PYTHONPATH=. python experiments/warp/exp_warp_gain_individual.py
274 PYTHONPATH=. python experiments/warp/exp_warp_age_low.py
275 PYTHONPATH=. python experiments/warp/exp_warp_age_low_holdout.py
276 PYTHONPATH=. python experiments/warp/exp_warp_drive.py --seeds 20
277 PYTHONPATH=. python experiments/warp/exp_warp_cross_session.py --layouts 10
```

### 278 C Full counterfactual sign map

### 279 References

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